A Joint Framework of Wavelet Filtering and Fast GSVT-LRSD Algorithm for SAR Narrowband Pulsed RFI Suppression

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***Abstract*— As the electromagnetic spectrum becomes increasingly crowded in recent years, synthetic aperture radar (SAR) is confronted with an escalating amount of radio-frequency interference (RFI). In civilian SAR satellite data, narrowband pulsed RFI (PRFI) is a prevalent interference type that significantly degrades the interpretability of SAR images. Among most approaches for suppressing PRFI, notch filtering methods face significant challenges in threshold selection of signal intensity. Conversely, low-rank and sparse decomposition (LRSD) algorithms, though free of threshold selection, often struggle to satisfy the required low-rank conditions. These limitations underscore the necessity of developing more robust interference suppression methods. In this article, we propose a joint framework of wavelet filtering and fast generalized singular value thresholding-based LRSD (FGSVT-LRSD) method to suppress narrowband PRFI in range–frequency and azimuth–time domain of SAR single-look complex (SLC) data. First, a wavelet domain notch filtering (WNF) method is employed to extract the strong spectral components that are primarily composed of PRFI in the 2-D range spectrum of SAR SLC data, while simultaneously protecting the spectrum of low-energy useful signals. Then, the FGSVT-LRSD method is applied to the extracted strong spectral components to efficiently separate the PRFI from the useful signals. By strategically integrating these two approaches, the proposed framework simultaneously exploits the high-intensity and low-rank properties of PRFI for more precise separation, significantly reduces the sensitivity of threshold selection in the WNF process and facilitates the low-rank conditions required by the FGSVT-LRSD method. Finally, the separated PRFI spectrum is subtracted from the original 2-D range spectrum, resulting in the RFI-suppressed SAR image. Experimental results based on both simulated and measured spaceborne SAR data will be presented to demonstrate that, compared with existing methods, the proposed approach exhibits superior PRFI suppression capabilities and effectively preserves useful signals.**

***Index Terms*— Low-rank and sparse decomposition (LRSD), narrowband interference suppression, pulsed radio-frequency interference (PRFI), synthetic aperture radar (SAR), wavelet filtering.**

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I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave remote sensing system known for its high-resolution, all-day, and all-weather capabilities, which has found extensive applications in the field of remote sensing [1], [2]. As a broadband microwave system, SAR shares frequency bands with artificial radio-frequency radiation sources [3], making it vulnerable to radio-frequency interference (RFI) and thus causing a decrease in its observation ability. In recent years, with the increasing amount of wireless devices, the electromagnetic spectrum has become increasingly crowded. This phenomenon has substantially increased the prevalence of RFI contamination in SAR systems, especially those operating in the L and C bands.

As one of the common forms, pulsed RFI (PRFI) originates from the pulsed signals emitted by ground radiation sources and is directly received by SAR receivers, causing aliasing between SAR echoes and PRFI [4]. In the SAR imaging process, although PRFIs are mismatched with the 2-D matched filter, their energy will diffuse throughout the image, resulting in a degradation of image quality. Consequently, there is an urgent need for effective methods to suppress PRFI during SAR operations.

In recent decades, there has been a continuous influx of studies to suppress RFI in SAR. These can be primarily categorized into three classifications: parametric methods, nonparametric methods, and semiparametric methods [5].

Among these methods, parametric methods employ a narrowband signal model to estimate the frequency and amplitude of RFI. Some iterative approaches, such as the iterative adaptive approach [6] and the gradual RELAX algorithm [7], have been proposed to estimate the frequency of RFI within a limited number of iterations. In contrast, the complex empirical mode decomposition method attempts to decompose narrowband RFI into multiple intrinsic-mode functions to achieve the separation of RFI and SAR signals [8]. Recently,

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signal estimation methods based on high-order polynomial estimation [9] or instantaneous frequency estimation [10] are proposed to attain superior RFI suppression results. However, the RFI suppression model differs from the conventional one used for estimating sinusoidal signal sources. As a matter of fact, the received signal comprises narrowband RFI, useful SAR signals, and noise. Consequently, direct estimation of sinusoidal signal parameters will result in the construction of an inaccurate RFI subspace, which in turn degrades the RFI suppression performance.

Apart from these, nonparametric methods are widely used due to their simplicity and efficiency, including adaptive filtering, notch filtering, and matrix decomposition methods [11], [12]. The main idea of adaptive filtering methods is to construct an adaptive filter within an appropriate domain to separate SAR signals from RFI. In [13], a time domain least mean square filter is employed to suppress multiple narrowband active interferences from NASA AirSAR data. Subsequently, an adaptive finite impulse response (FIR) filter is constructed in [14] to suppress RFI. Similar to the adaptive FIR filter, a method such as singular value spectrum analysis is proposed in [15]. Other types of nonparametric methods, for example, matrix decomposition methods, such as principal component analysis (PCA) [16], eigen-subspace projection [17], and independent component analysis [18], [19], are suitable for powerful RFI suppression, but suffer from great computational costs. For efficiency, notch filtering suppresses RFI signals by setting them to 0, with the frequency domain notch filtering (FNF) method being one of the most effective approaches [20]. The simplicity and robustness of the FNF method make it widely used in RFI suppression applications such as the ALOS-1 PALSAR SAR system [21]. Recently, a time domain notch filtering method is put forward to suppress PRFI in SAR raw data [22], which employs a missing data recovery technique to fill the notched gaps. Due to the difficulty of acquiring SAR raw data, the subband spectral cancellation method [23], the block subspace filtering method [24], and range–frequency domain iterative notch filtering method [25], [26], perform RFI suppression on SAR single-look complex (SLC) image data, which are easier to implement in the real world. Although nonparametric methods offer the benefits of simplicity and high efficiency, they inevitably result in some loss of useful signals, which in turn reduces the matched filtering gain during the SAR imaging process [27].

In recent years, semiparametric methods have been applied to suppress RFI using a low-rank and sparse decomposition (LRSD) model [11]. These methods utilize the low-rank characteristic of narrowband RFI and the sparsity of useful signals within the synthetic aperture time [28], thereby converting the complex signal separation problems into convex optimization problems that involve hyperparameters. The robust PCA (RPCA) algorithm was first proposed in [29] and [30] to suppress narrowband RFI, utilizing nuclear norm for rank constraint. In [31], a reweighted nuclear norm minimization (NNM) method is employed to mitigate the drawbacks of NNM. Thereafter, a matrix factorization technique is used in [32] to accelerate the aforementioned reweighted NNM calculation. For a better representation of the low-rank property of narrowband RFI, the block Bayesian method [33] and a tensor recovery method [34] are introduced, realizing commendable suppression performance. However, the computational burden of the above algorithms is extremely high.

Current research mainly focuses on RFI suppression for level-0 raw data, with limited attention given to level-1 SLC images despite their accessibility. The development of RFI suppression on level-1 data begins with the introduction of a posteriori filtering algorithm [35], followed by advancements such as a blockwise RPCA algorithm designed to mitigate mutual interference in spaceborne SAR systems [36]. Recent progress includes a generalized singular value thresholding-based LRSD (GSVT-LRSD) algorithm that provides a new LRSD approach for SLC images [37] and an adaptive method combining FNF with a semiparametric method for robust scene recovery under strong RFI conditions [38]. However, the threshold iteration strategy of the method in [38] needs to be modified for different scenarios.

Most notch filtering methods face significant challenges in threshold selection, while LRSD methods often struggle to satisfy the required low-rank conditions in measured SAR data. In this article, we propose a joint framework of wavelet domain notch filtering (WNF) and a fast GSVT-LRSD (FGSVT-LRSD) algorithm to suppress narrowband PRFI in SAR level-1 SLC image data. First, a wavelet decomposition is employed on the range–frequency and azimuth–time domain of the SAR SLC image data. Then, by detecting the wavelet coefficients exceeding the threshold in the wavelet domain, the WNF method is used to extract the spectral components representing PRFI in the 2-D range spectrum, while simultaneously protecting the spectrum of useful signals. Furthermore, the FGSVT-LRSD method is applied to the extracted spectral components to separate the narrowband PRFI from the useful signals. Finally, we subtract the separated PRFI spectrum from the original range spectrum to obtain the RFI-suppressed SAR image.

The remainder of this article is organized as follows. In Section II, the narrowband PRFI signal model and characteristics are introduced. Then, Section III elaborates on the proposed joint framework of WNF and FGSVT-LRSD (WNF-FGSVT-LRSD). In Section IV, the performance and analysis of the experimental results on simulation and measured SAR data are given. At last, Section V concludes this article.

II. Signal Model and Problem Formulation

A. Signal Model of PRFI

Most of the PRFIs encountered by SAR systems originate from high-power ground-based radiation sources such as radars and communication devices. These radiation sources periodically emit strong pulsed signals. If they are located within the SAR observation area, the radio-frequency pulsed signals will reach the SAR receiver in the form of direct wave signals and mix with the SAR echo signals in the SAR receiving window [39]. In general, the timing diagrams for the transmission and reception between SAR and PRFI sources are illustrated in Fig. 1.

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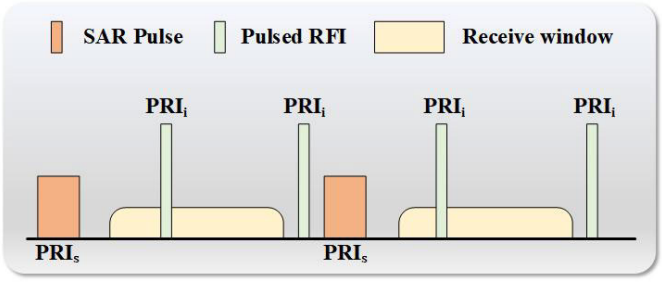


Fig. 1. Transmitting and receiving timing diagrams of SAR echo and PRFI.

The mathematical form of this mixed signal is as follows:

$$\mathcal{M}(t_r, t_a) = \mathcal{S}(t_r, t_a) + \mathcal{J}(t_r, t_a) + \mathcal{N}(t_r, t_a) \tag{1}$$

where t_r and t_a denote the range–time and the azimuth–time, respectively; and $\mathcal{S}(t_r, t_a)$, $\mathcal{J}(t_r, t_a)$, and $\mathcal{N}(t_r, t_a)$ signify the useful signal, interference, and noise [40], respectively. In addition, $\mathcal{M}(t_r, t_a)$ represents a mixture of the aforementioned signals. The pulsed point frequency (PPF) interference signals can be modeled as a bulk of sinusoidal signals as follows:

$$\mathcal{J}_{\text{PPF}}(t_r, t_a) = \sum_{p=1}^{P} A_p(t_a) \cdot \text{rect}\left(\frac{t_r}{T_{t_a}}\right) \cdot \text{rect}\left(\frac{t_r - t_{r_0}}{T_0}\right) \cdot \exp\{j\left(2\pi f_p t_r + \varphi_p\right)\} \tag{2}$$

where T_{t_a} denotes the SAR signal receiving duration, P is the number of PRFI components, whereas $A_p(t_a)$, f_p, and φ_p denote the amplitude, the center frequency, and the initial phase of the pth pulsed NBI component, respectively. t_{r_0} is the central time of the PRFI. T_0 represents the pulsewidth of PRFI.

The pulsed narrowband linear frequency modulation (PNLFM) interference can be expressed as

$$\mathcal{J}_{\text{PNLFM}}(t_r, t_a) = \sum_{p=1}^{P} A_p(t_a) \cdot \text{rect}\left(\frac{t_r}{T_{t_a}}\right) \cdot \text{rect}\left(\frac{t_r - t_{r_0}}{T_0}\right) \cdot \exp\left\{j\left(2\pi f_p t_r + \pi K_p t_r^2\right)\right\} \tag{3}$$

where K_p represents the chirp rate of the PNLFM component within the pth pulse.

B. Problem Formulation

In the range–frequency and azimuth–time domain, 2-D narrowband PRFI exhibits relatively stable frequencies along the azimuth–time axis. This narrowband PRFI appears as intermittent parallel straight lines, visualized in Fig. 2. Evidently, for the 2-D PRFI matrix, it possesses the low-rank property in the azimuth–time direction, whereas the SAR echo is characterized by its sparsity. Based on this characteristic, we can separate and extract the PRFI components from the observed SAR data by addressing the following LRSD problem:

$$\min_{\mathbf{L},\mathbf{S}} \text{rank}(\mathbf{L}) + \lambda\|\mathbf{S}\|_0 \quad \text{s.t. } \|\mathbf{Y} - \mathbf{L} - \mathbf{S}\|_{\text{F}}^2 < \delta \tag{4}$$

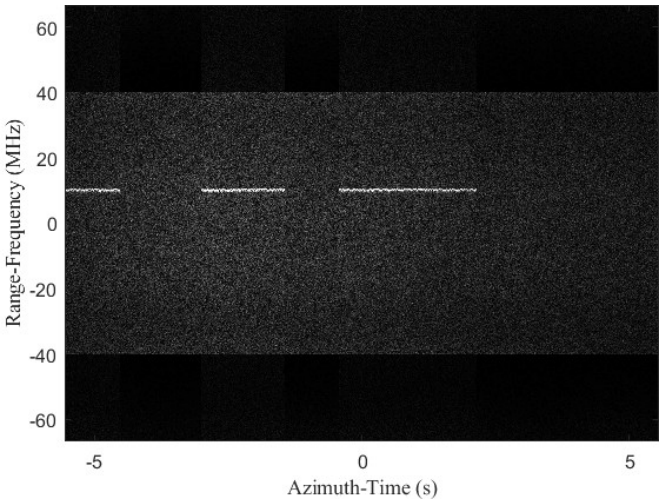


Fig. 2. Narrowband PRFI contaminated SAR image in range–frequency and azimuth–time domain.

where rank$(\cdot)$ presents the rank calculation operation, $\mathbf{L}$ denotes the low-rank part of the mixed signal matrix, $\mathbf{S}$ denotes the sparse part that represents SAR signal matrix, $\mathbf{Y}$ is the Lagrangian multiplier, and λ and δ denote the hyperparameters.

The aforementioned optimization problem is an NP-hard one, making it extremely challenging to solve. Consequently, convex relaxation becomes necessary. If we approximate the rank function via the nuclear norm and replace the ℓ_0 norm with the ℓ_1 norm, the above optimization problem will be transformed into a classic RPCA problem

$$\min_{\mathbf{L},\mathbf{S}} \|\mathbf{L}\|_* + \lambda\|\mathbf{S}\|_1 \quad \text{s.t. } \|\mathbf{Y} - \mathbf{L} - \mathbf{S}\|_{\text{F}}^2 < \delta. \tag{5}$$

By alternately iterating to solve the matrices $\mathbf{L}$ and $\mathbf{S}$, we can obtain the decomposed low-rank matrix $\mathbf{L}$ and sparse matrix $\mathbf{S}$, thus completing the separation of PRFI and SAR echoes. During the solution process, to address the NNM problem, we typically employ the singular value thresholding (SVT) operator to extract the low-rank part of the matrix

$$\text{SVT}(\varrho(\mathbf{Y})) = \{(\varrho_i(\mathbf{Y}) - \zeta)_+\}_{i=1}^{\alpha} \tag{6}$$

in which $\varrho_i(\mathbf{Y})$ represents the ith singular value of $\mathbf{Y}$, ζ is the threshold, and $\mathbf{Y}$ is a $\alpha \times \beta$-dimensional matrix to be calculated. Since the SVT factor in (6) applies the same threshold to all singular values, it will lead to excessive penalization of large singular values [41]. However, large singular values correspond to the main components of the matrix, namely, the low-rank part, while small singular values correspond to the sparse part. As a result, after the SVT calculation, part of the PRFI energy remains in the SAR echoes, causing a decline in PRFI suppression performance. Therefore, the SVT operator needs to be improved, and the specific details will be elaborated in Section III.

III. Proposed Method

When the RFI presents as pulsed signals or fails to fully disrupt all azimuth–time of SAR echoes, the existing time or

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FNF methods sustain substantial loss of useful signals, while the LRSD methods suffer from excessively long computation times. To tackle these problems, we propose our WNF-FGSVT-LRSD method for suppressing narrowband PRFI in SAR. This method starts by applying a range fast Fourier transform (FFT) to the SAR SLC data contaminated by PRFI, transforming it into the range–frequency and azimuth–time domain. Then, out-of-band interference is removed based on the proportional relationship between the bandwidth and the sampling rate. Subsequently, wavelet decomposition is performed on the truncated spectrum to obtain the low-frequency trend features of the spectrum. Through adaptive threshold selection, the components with relatively high energy are screened and extracted. For the extracted spectral components, an FGSVT-LRSD algorithm is carried out. As a result, the low-rank part $\mathbf{L}$ (representing the narrowband interference) and the sparse part $\mathbf{S}$ (corresponding to the echo spectrum) are separated. By subtracting $\mathbf{L}$ from the original 2-D range spectrum and then performing an inverse FFT (IFFT) to transform the RFI-suppressed range spectrum into the echo or the complex image, interference suppression can be achieved. The proposed method provides rapid PRFI suppression with minimal loss of useful signal. The above-mentioned process is illustrated in the flowchart (see Fig. 3).

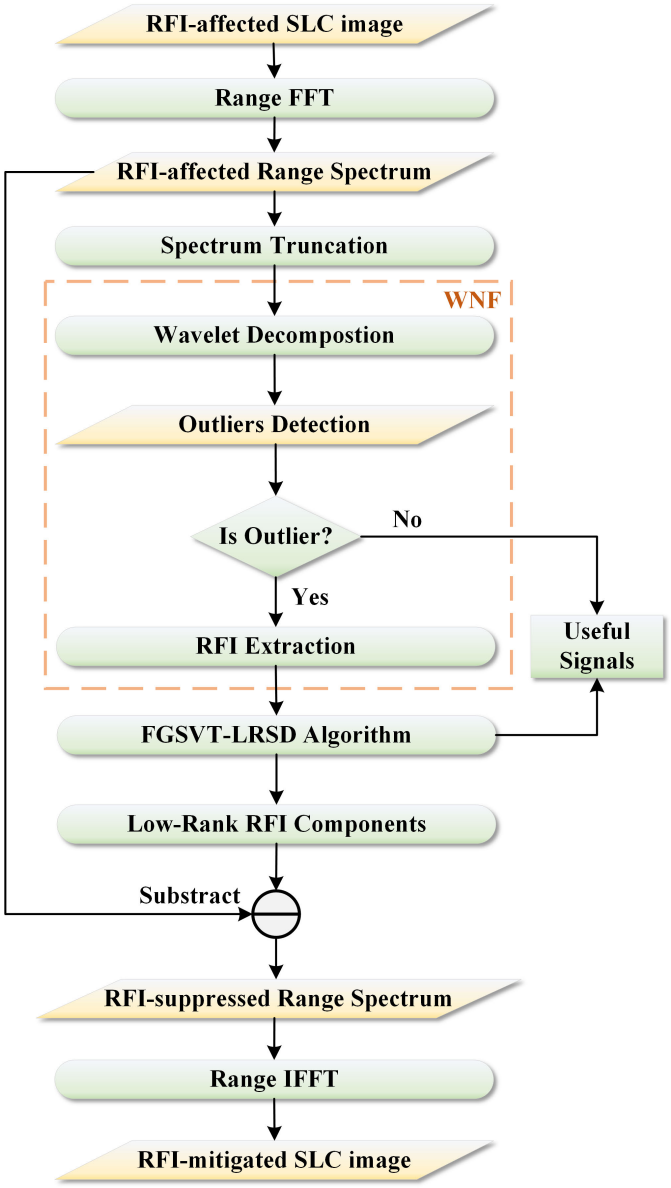


Fig. 3. Flowchart of the proposed WNF-FGSVT-LRSD method.

A. Spectrum Truncation

First, SAR SLC data is transformed into a range–frequency spectrum by the range FFT. To ensure alias-free signal acquisition in radar applications, the Nyquist criterion mandates a sampling rate f_s that exceeds 1.2 times the operational bandwidth B. In most cases, the frequency components beyond the operational band exhibit negligible energy levels. However, PRFI can occur in these peripheral spectral regions and may be detected by the SAR receiver. Therefore, spectral truncation can effectively eliminate the impact of out-of-band PRFI. Moreover, the truncation of these nonessential spectral components enables effective dimensionality reduction. Assuming a 2-D range spectrum with a size of N_r along the range–frequency axis and N_a along the azimuth–time axis, the M spectral points can be retained from the initial range spectrum by truncation, where $M = N_r \cdot (1 - (B/f_s))$. This optimized data representation substantially alleviates the computational burden inherent in the FGSVT-LRSD method detailed in the following section, particularly regarding matrix operations and iterative calculations. In addition, the original signal dimensions can be recovered through zero padding, preserving system compatibility. Quantitative evaluations of processing efficiency and temporal performance metrics are systematically compared in Section IV.

B. WNF Process

The wavelet transform enables localized time–frequency analysis through scaling and shifting operations, offering enhanced capability for transient feature extraction compared with conventional Fourier methods. Unlike the infinite trigonometric basis functions in Fourier analysis, wavelet transforms utilize finite decaying basis functions that approximate signals more effectively with abrupt variations. This characteristic proves particularly valuable for processing SAR signals contaminated by RFI, as the two components exhibit distinct spectral properties. While SAR echoes follow complex Gaussian distributions in the range–frequency domain, RFI manifests as concentrated narrowband components, typically representing aggregated amplitude or frequency-modulated signals.

Our WNF approach leverages this spectral distinction to separate the impulsive RFI components from the relatively smooth SAR echoes. First, a range FFT operation is performed on the RFI-contaminated SLC image to acquire the 2-D range spectrum

$$\mathbf{S}_a(f_r, t_a) = \mathrm{FFT}(\mathbf{I}(t_r, t_a)) \tag{7}$$

where $\mathbf{S}_a$ represents the RFI-affected range spectrum, $\mathbf{I}$ denotes the RFI-affected SLC image, f_r is the

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range–frequency axis, and t_r and t_a represent range–time and azimuth–time axis, respectively.

After the spectrum truncation and zero padding mentioned in the paragraph above, the truncated range spectrum can be represented as $\mathbf{S}_T$. Given that the Daubechies wavelet exhibits superior frequency selection characteristics [42], this method performs an iterative C-level wavelet decomposition for each azimuth–time of the truncated spectrum. The wavelet decomposition process for the azimuth–time t_{a_i} at the cth level can be mathematically expressed as

$$\mathbf{W}_c(w_r, t_a) = \mathcal{D}_c(|\mathbf{S}_T(f_r, t_a)|)_{t_a=t_{a_i}} \tag{8}$$

where $|\cdot|$ denotes the operation of taking the absolute value, $\mathbf{S}_T(f_r, t_a) \in \mathbb{C}^{M\times N_a}$ denotes the truncated range spectrum, $\mathcal{D}_c$ represents the Daubechies wavelet decomposition operator with $c = 1, 2, \ldots, C$, and $\mathbf{W}_c(w_r, t_a) \in \mathbb{R}^{M'\times N_a}$ denotes the wavelet coefficients after the cth level wavelet decomposition and $M' \approx M/2^c$. Before taking the absolute value, the phase Φ_s of $\mathbf{S}_T(f_r, t_a)$ needs to be calculated and retained.

Then, an outlier detection is performed in the wavelet domain to identify RFI-contaminated frequency components. The inherent decorrelation capability of wavelet transforms induces distinct energy distribution patterns: signal energy concentrates in dominant wavelet coefficients, while noise energy disperses uniformly across the entire wavelet domain. Thus, RFI-contaminated components exhibit substantially larger coefficients compared with noise-like echoes. This disparity enables threshold-based separation. To establish robust threshold criteria, we employ the interquartile range (IQR) method [43], which demonstrates resilience to extreme values in non-Gaussian distributions. The expression for IQR is as follows:

$$\mathrm{IQR} = Q_3 - Q_1 \tag{9}$$

where Q_3 is the third quartile, and 75% of the wavelet coefficients lie below this value. Q_1 denotes the first quartile, indicating that 25% of the wavelet coefficients fall below this value. IQR represents the middle 50% of data spread. Furthermore, the upper and lower boundaries for determining outliers can be expressed by the following equations:

$$B_u = Q_3 + \gamma \times \mathrm{IQR} \tag{10}$$

$$B_l = Q_1 - \gamma \times \mathrm{IQR} \tag{11}$$

where B_u denotes the upper bound, B_l represents the lower bound, and γ is the threshold factor which is adjustable based on the confidence level and is typically set to 1.5. Wavelet coefficients that exceed these boundaries are identified as outliers and preserved. Here, taking the upper bound as an example, the equation for the hard threshold operation can be expressed as follows:

$$\hat{\mathbf{W}}_c(w_r, t_a) = \begin{cases} 0, & |\mathbf{W}_c(w_r, t_a)| \le B_u \\ \mathbf{W}_c(w_r, t_a), & |\mathbf{W}_c(w_r, t_a)| > B_u \end{cases} \tag{12}$$

in which $\mathbf{W}_c(w_r, t_a)$ represents the wavelet coefficient of c-level wavelet decomposition and B_u denotes the threshold calculated by the IQR method.

This IQR-derived hard thresholding ensures reliable detection of anomalous wavelet components while preserving the Gaussian distributed echo characteristics inherent to SAR signals. Finally, we perform wavelet reconstruction on the hard-thresholded wavelet coefficients at all levels to obtain the magnitude of the range spectrum. The wavelet reconstruction process for the azimuth–time t_{a_j} at the cth level can be expressed as

$$\mathbf{S}_w(f_r, t_a) = \mathcal{R}_c\left(\left|\hat{\mathbf{W}}_c(w_r, t_a)\right|\right)_{t_a=t_{a_j}} \tag{13}$$

where $\mathcal{R}_c$ is the wavelet reconstruction operator and $\mathbf{S}_w(f_r, t_a)$ is the amplitude of reconstructed range spectrum.

Subsequently, the previously preserved phase information Φ_s of the range spectrum is integrated to reconstruct the complex-valued range spectrum as follows:

$$\mathbf{S}_r(f_r, t_a) = \mathbf{S}_w(f_r, t_a) \odot e^{j\Phi_s} \tag{14}$$

where $\mathbf{S}_r(f_r, t_a)$ represents the complex-valued range spectrum after WNF and $\odot$ is the elementwise multiplication operator. As a result, the proposed WNF process achieves effective PRFI extraction while minimizing distortion of the desired characteristics of SAR signals.

C. FGSVT-LRSD Algorithm

Since low rankness is an extension of sparsity to the singular values of a matrix, the GSVT algorithm [44] for the LRSD problem in (5) enhances rank function characterization by extending the nonconvex surrogate function of the ℓ_0-norm to matrix singular values. This extension achieves higher precision in low-rank matrix approximation and the elimination of the uniform treatment of singular values inherent to conventional rank functions. The resulting optimization model, incorporating the nonconvex surrogate function, is formulated as

$$\begin{aligned} &\min \sum_{i=1}^{m} g(\varrho_i(\mathbf{L})) + \lambda\|\mathbf{S}\|_1 \\ &\text{s.t. } \mathbf{X} = \mathbf{L} + \mathbf{S} \end{aligned} \tag{15}$$

where $\mathbf{X}$ represents the 2-D range spectrum of SAR signals and $\mathbf{L}$ and $\mathbf{S}$ stand for the low-rank and sparse components of $\mathbf{X}$, respectively. $g(x)\colon \mathbb{R}^+ \to \mathbb{R}^+$ denotes an arbitrary nonconvex surrogate function, the gradient ∇g is convex, when $g(x) = x$, $\sum_i^{\min(M,N_a)} g(\varrho_i(\mathbf{L}))$ is equal to the nuclear norm [37].

Different from the soft thresholding function (6) used in RPCA, the GSVT-LRSD method presents a GSVT factor $\mathrm{Prox}_g^{\varrho}(\cdot)$ to solve the nonconvex low-rank minimization problem

$$\mathrm{Prox}_g^{\varrho}(\mathbf{D}) = \arg\min_{\mathbf{L}} \sum_{i=1}^{m} g(\varrho_i(\mathbf{L})) + \frac{1}{2}\|\mathbf{L} - \mathbf{D}\|_{\mathrm{F}}^2. \tag{16}$$

Let $\varepsilon_1^* \ge \varepsilon_2^* \ge \cdots \ge \varepsilon_m^* \ge 0$ denotes the first m largest singular values of matrix $\mathbf{L}$, where each $\varepsilon_i^* \in \mathrm{Prox}_g(\varrho_i(\mathbf{D}))$. Based on these definitions, (16) can be represented as

$$\arg\min_{\varepsilon_1^*\ge\cdots\ge\varepsilon_m^*\ge 0} \sum_{i=1}^{m} \left(g(\varepsilon_i^*) + \frac{1}{2}\left(\varepsilon_i^* - \varrho_i(\mathbf{D})\right)^2 \right). \tag{17}$$

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TABLE I
NONFULL-RANK SVD

Algorithm 1 A randomized algorithm for non-full-rank SVD

Inputs: $\mathbf{D}^{M\times N_a}$, $K \le \min\{M, N_a\}$
Initialization: $\Omega^{N_a\times K}$
Update: $\mathbf{M} = \mathbf{D}\Omega$
Update: $\mathbf{Q} = \mathrm{qr}(\mathbf{M})$[1]
Update: $\mathbf{B} = \mathbf{Q}^{\mathbf{T}}\mathbf{D}$
Update: $[\mathbf{U_B}, \mathbf{\Sigma_B}, \mathbf{V_B}] = \mathrm{SVD}(\mathbf{B})$
Update: $\mathbf{U_B} = \mathbf{Q}\mathbf{U_B}$
Outputs: $\mathbf{U_X} = \mathbf{U_B}(:, 1:K), \mathbf{\Sigma_X} = \mathbf{\Sigma_B}(:, 1:K)$, $\mathbf{V_X} = \mathbf{V_B}(:, 1:K)$

[1] Here qr denotes the orthogonal triangle decomposition operation.

TABLE II
FAST GSVT ALGORITHM

Algorithm 2 The procedure of FGSVT calculation

Inputs: $\mathbf{X}^{M\times N_a}$, $\mathbf{S}$, $\mathbf{Y}$, μ^k, RL^k
Initialization: $k = 0$, $\mathbf{L}_0 = 0$, $\mathbf{S}_0 = 0$, $\mu_0 = \frac{1.25}{\|\mathbf{X}\|_2}$, $\mathbf{Y_0} = \frac{\mathbf{X}}{\max(\|\mathbf{X}\|_2, \sqrt{MN_a}\|\mathbf{X}\|_\infty)}$, $RL_0 = \min(M, N_a)$
Update:
$[\mathbf{U}_{RL^k}, \mathbf{\Sigma}_{RL^k}, \mathbf{V}_{RL^k}] \leftarrow \mathrm{rSVD}(\mathbf{X} - \mathbf{S}^k + \mathbf{Y}^k, RL^k)$
$\hat{x_i} \leftarrow \max\{x_i | \nabla f(x_i) = 0, 0 < i \le RL^k, x_i \in \mathbf{\Sigma}_{RL^k}\}$[1]
$RL^{k+1} \leftarrow i$
$\mathbf{L}^{k+1} \leftarrow \mathbf{U}_{RL^{k+1}} \cdot \mathbf{\Sigma}_{RL^{k+1}} \cdot \mathbf{V}^T_{RL^{k+1}}$
Outputs: $\mathbf{L}^{k+1}, RL^{k+1}$

[1] Here $\hat{x_i}$ is find through a general solver proposed in [41] and $f(x) = g(x) + \frac{1}{2}(x-d)^2$.

Then, an optimal solution to (16) is $\mathbf{L}^* = \mathbf{U}\,\mathrm{Diag}(\boldsymbol{\varepsilon}^*)\mathbf{V}^{\mathrm{T}}$. More generally, (16) can be transformed into

$$\mathrm{Prox}_g^{\varrho}(\mathbf{D}) = \mathbf{U}\mathrm{Diag}(\mathrm{Prox}_g(\varrho(\mathbf{D})))\mathbf{V}^{\mathrm{T}} \tag{18}$$

where $\mathbf{D} = \mathbf{U}\mathrm{Diag}(\varrho(\mathbf{D}))\mathbf{V}^{\mathrm{T}}$ is the singular value decomposition (SVD) of $\mathbf{D}$ and $\mathrm{Diag}(\cdot)$ denotes the diagonal matrix operator [45].

For each $d = \varrho_i(\mathbf{D})$ of an arbitrary matrix $\mathbf{D}$, (17) can be transformed into solving the following equation:

$$\mathrm{Prox}_g(d) = \arg\min_{x\ge 0} f(x) = g(x) + \frac{1}{2}(x-d)^2. \tag{19}$$

In [41], a general solver is proposed to find $\mathrm{Prox}_g(d)$ for a certain type of g. The solutions to (19) lie in $x = 0$ or the local points $\{x|\nabla f(x) = \nabla g(x) + x - d = 0\}$. Among the equations above, we find that the SVD operation contained in (18) has the highest computational complexity in the process. Since it has been previously proven that narrowband interference has low-rank characteristics, when calculating the $\mathrm{Prox}_g(\cdot)$ operator in the process of solving the low-rank component, instead of performing a full-rank SVD, we can use a nonfull-rank SVD of rank K [where $K \ll \min(M, N_a)$] for approximate representation, so as to accelerate the algorithm.

The Johnson–Lindenstrauss lemma establishes that high-dimensional data can be efficiently embedded into a lower dimensional space via random projection [46]. This dimensionality reduction is achieved through a randomly constructed projection matrix Ω that approximately preserves the relative distances with high probability. Notably, this lemma provides a computationally efficient alternative to exact SVD, as it enables the approximation of nonfull-rank SVD operations without requiring complete matrix factorization. The pseudocode of nonfull-rank SVD using random projection is shown in Table I.

The augmented Lagrangian function of the model in (15) is constructed as follows:

$$\Gamma(\mathbf{L}, \mathbf{S}, \mathbf{Y}, \mu) = \sum_{i=1}^{K} g(\varrho_i(\mathbf{L})) + \lambda\|\mathbf{S}\|_1 - \langle \mathbf{Y}, \mathbf{L} + \mathbf{S} - \mathbf{X}\rangle + \frac{\mu}{2}\|\mathbf{L} + \mathbf{S} - \mathbf{X}\|_{\mathrm{F}}^2 \tag{20}$$

where μ is a hyperparameter, $\mathbf{Y}$ denotes the Lagrangian multiplier, and $\langle\cdot\rangle$ is the matrix inner product. To achieve successful separation of interference components from useful signals, the alternating direction method of multipliers (ADMM) framework provides an effective solution

$$\begin{aligned}\mathbf{L}^{k+1} &= \arg\min_{\mathbf{L}} \Gamma_a(\mathbf{L}, \mathbf{S}^k, \mathbf{Y}^k)\\ \mathbf{S}^{k+1} &= \arg\min_{\mathbf{S}} \Gamma_a(\mathbf{L}^{k+1}, \mathbf{S}, \mathbf{Y}^k)\\ \mathbf{Y}^{k+1} &= \mathbf{Y}^k + \rho\nabla h(\mathbf{Y})\\ &= \mathbf{Y}^k + \rho(\mathbf{L}^{k+1} + \mathbf{S}^{k+1} - \mathbf{X})\end{aligned} \tag{21}$$

where $h(\cdot)$ is the dual function of the Lagrangian function.

According to the above equations, the low-rank matrix $\mathbf{L}$ can be updated as follows:

$$\begin{aligned}\mathbf{L}^{k+1} &= \arg\min_{\mathbf{L}} \mathcal{L}(\mathbf{L}, \mathbf{S}^k, \mathbf{Y}^k, \mu^k)\\ &= \mathrm{Prox}_{\frac{g}{\mu^k}}^{\varrho}\left(\mathbf{X} - \mathbf{S}^k + \frac{\mathbf{Y}^k}{\mu^k}\right)\\ &= \mathbf{U}^k\mathrm{Diag}\left\{\mathrm{Prox}_{\frac{g}{\mu^k}}\left(\varrho\left(\mathbf{X} - \mathbf{S}^k + \frac{\mathbf{Y}^k}{\mu^k}\right)\right)\right\}(\mathbf{V}^k)^{\mathrm{T}}\end{aligned} \tag{22}$$

where $\mathbf{U}^k$ and $\mathbf{V}^k$ are calculated via $\mathbf{X} - \mathbf{S}^k + (\mathbf{Y^k}/\mu^k)$.

Since updating $\mathbf{L}$ requires performing GSVT at each iteration, which involves SVD, the computational cost can be significant, especially when the true rank RL of the low-rank matrix $\mathbf{L}$ is unknown. In order to accelerate the GSVT algorithm using randomized SVD (rSVD), we modify the procedure of the GSVT calculation and present our proposed FGSVT algorithm. The pseudocode of the proposed FGSVT is shown in Table II.

After the update of $\mathbf{L}$, we fix $\mathbf{L}$ and $\mathbf{Y}$ and update $\mathbf{S}$, as shown in the following equation:

$$\begin{aligned}\mathbf{S}^{k+1} &= \arg\min_{\mathbf{S}} \mathcal{L}(\mathbf{L}^{k+1}, \mathbf{S}, \mathbf{Y}^k, \mu^k)\\ &= \arg\min_{\mathbf{S}} \frac{\lambda}{\mu^k}\|\mathbf{S}\|_1 + \frac{1}{2}\left\|\mathbf{S} - \left(\mathbf{X} - \mathbf{L}^{k+1} + \frac{\mathbf{Y}^k}{\mu^k}\right)\right\|_{\mathrm{F}}^2\\ &= \mathrm{So}_{\frac{\lambda}{\mu^k}}\left(\mathbf{X} - \mathbf{L}^{k+1} + \frac{\mathbf{Y}^k}{\mu^k}\right)\end{aligned} \tag{23}$$

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where $\mathrm{So}_{(\lambda/\mu)}(\mathbf{D}) = \max(|\mathbf{D}| - (\lambda/\mu), 0) \cdot \mathrm{sign}(\mathbf{D})$, $(\lambda/\mu) > 0$, and sign$(\cdot)$ stands for sign function. Finally, we update the Lagrange multiplier $\mathbf{Y}$ and the hyperparameter μ

$$\mathbf{Y}^{k+1} = \mathbf{Y}^k - \mu^k\left(\mathbf{L}^{k+1} + \mathbf{S}^{k+1} - \mathbf{X}\right) \tag{24}$$

$$\mu^{k+1} = \min(\rho\mu^k, \mu_{\max}) \tag{25}$$

where ρ is a hyperparameter and $\rho > 1$.

Upon convergence through iterative optimization of the described framework, the algorithm achieves accurate extraction of narrowband PRFI signals, which are mathematically characterized as the low-rank components within the range spectrum

$$\mathbf{J}_{\mathrm{PRFI}}(f_r, t_a) = \mathbf{L}^k. \tag{26}$$

This process effectively isolates PRFI from the desired signal components, demonstrating robust performance in interference suppression. The RFI-suppressed result is obtained by removing narrowband PRFI signals in the range spectrum. This process can be mathematically represented as

$$\mathbf{S}_m(f_r, t_a) = \mathbf{S}_r(f_r, t_a) - \mathbf{J}_{\mathrm{PRFI}}(f_r, t_a) \tag{27}$$

where $\mathbf{S}_m$ denotes the RFI-suppressed range spectrum. Then, by applying the IFFT along the range direction to the zero-padded RFI-suppressed range spectrum, the RFI-free SAR SLC image data can be obtained

$$\mathbf{I}_m(t_r, t_a) = \mathrm{IFFT}(\mathbf{S}_m(f_r, t_a)) \tag{28}$$

where $\mathbf{I}_m$ is the RFI-free SLC image.

IV. Experiment Results and Analysis

In this section, the PRFI suppression performance of the proposed WNF-FGSVT-LRSD method, along with several state-of-the-art methods, is analyzed both qualitatively and quantitatively through experiments utilizing simulated and measured SAR SLC data. Specifically, PRFI suppression experiments are conducted under simulated SAR data in the case of different interference-to-signal ratios (ISRs). Then, the PRFI suppression performances are analyzed through root mean square error (RMSE). The superiority of the proposed method over existing RFI suppression techniques is validated through quantitative evaluation of three different SAR SLC images. Particularly, the average gradient (AG) and equivalent number of looks (ENL) metrics are computed for the RFI-suppressed images, demonstrating improvements in image quality. At last, to elucidate the connection between the two methods within the joint framework, we conduct discussions and analysis. The experiments are carried out on an HP workstation equipped with an Intel i7-10700 CPU and 64G of computer memory, which is located in the school laboratory.

A. Simulation Experiment Results

In the simulation experiments, we utilized a real RFI-free SAR scene. The contaminated SAR scenario is generated by adding simulated PRFI signals into SAR echoes, based on the system parameters of the C-band GF-3 SAR satellite and the interference characteristics detailed in Table III. Specifically,

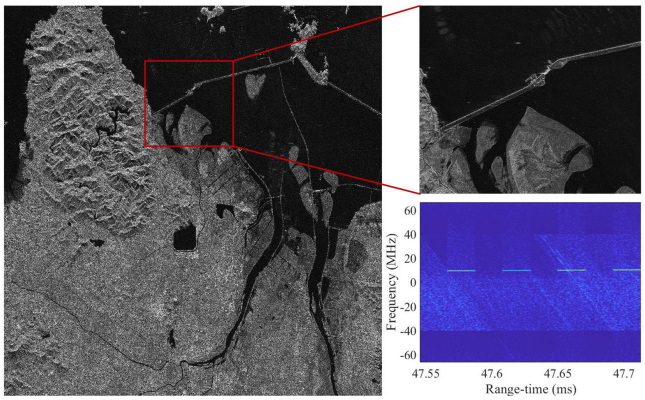


Fig. 4. GF-3 SAR image and the narrowband RFI used in the simulation experiments.

TABLE III
Parameters of Simulation

Parameters	Values
Center Frequency	5.400GHz
Sampling Frequency	133.3MHz
Platform Velocity	7128m/s
Center Slant Range	7126km
System Bandwidth	80MHz
Pulse Width	55μs
PRF	1948Hz
Center Frequency of RFI	5.402GHz
Bandwidth of RFI	1MHz
Pulse Width of RFI	21μs
Interference to Signal Ratio	[0,20]dB

the RFI is modeled as a narrowband PRFI with a 1-MHz bandwidth, and the ISR is systematically varied from 0 to 20 dB to evaluate the robustness of the proposed method across different contamination levels. Due to limited computational resources, the simulation experiments are conducted on a representative slice extracted from the full scenario. The SLC image of GF-3 SAR data, the selected slice of the scene, and the time–frequency spectrum of simulated narrowband PRFI contaminated SAR data are illustrated in Fig. 4.

Narrowband PRFI can be effectively represented as sinusoidal components, introducing systematic noise artifacts throughout the image. To quantitatively evaluate PRFI suppression performance, we utilize the RMSE metric, which compares the reconstructed SAR image against the ground truth image. RMSE, a widely adopted quantitative measure in image quality assessment [47], is calculated as follows:

$$\mathrm{RMSE}(\mathbf{I}, \mathbf{I}_m) = \frac{\|\mathbf{I} - \mathbf{I}_m\|_{\mathrm{F}}}{\|\mathbf{I}\|_{\mathrm{F}}} \tag{29}$$

where $\mathbf{I}$ denotes the original SLC image and $\mathbf{I}_m$ denotes the recovered SLC image. The numerator in (29) captures pixelwise deviations between the original SAR data and its reconstructed version, while the denominator corresponds to the Frobenius norm of the ground truth image. Since PRFI contamination behaves similar to structured noise, the RMSE

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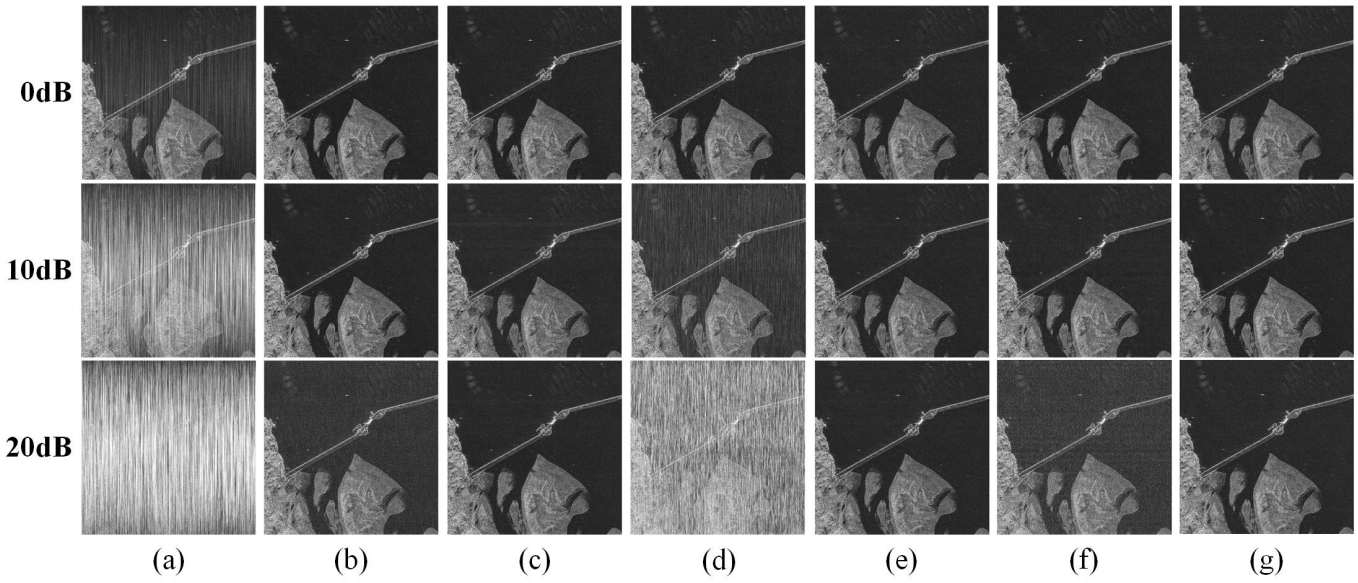


Fig. 5. Experimental results for simulated SAR data. (a) RFI-affected SAR image. Suppression outcome of (b) FNF, (c) PCA, (d) RPCA, (e) WNF, (f) GSVT-LRSD, and (g) proposed method.

metric inherently reflects two critical aspects: suppression of interference components and preservation of useful signal content. As a fidelity metric, a lower RMSE value directly indicates better PRFI suppression quality.

To validate the effectiveness of the proposed method, we conduct comprehensive simulations that compare its performance with five current methods, including FNF, PCA, RPCA, WNF, and GSVT-LRSD, and across varying ISR. The suppression performance is quantitatively evaluated using RMSE under diverse contamination levels, which enables a rigorous assessment of interference mitigation capability. In particular, the WNF and GSVT-LRSD methods are used, respectively, in the simulation experiments to verify the superiority of the proposed method in comparison to one of its modules. The generated RFI-contaminated scenarios and RFI-suppressed images are illustrated in Fig. 5.

As illustrated in Fig. 5(a), the degradation of SAR image quality intensifies with increasing ISR levels, culminating in complete obscuration of ground targets at an ISR of 20 dB. Analysis of the suppression results in Fig. 5(b)–(g) reveals a performance divergence of different methods at high ISR levels. When the ISR is at 0 dB, all methods exhibit effective PRFI suppression and a notable improvement in image quality. However, as the ISR increases to 10 dB, RPCA demonstrates limited suppression efficacy, resulting in substantial residual interference, while PCA fails to eliminate stripe-like artifacts in regions such as sea surfaces. When the ISR is at 20 dB, RPCA proves ineffective for RFI suppression and target recovery. Moreover, although FNF and GSVT-LRSD achieve relatively better suppression, their processed images still contain noise-like artifacts that degrade texture details and overall image quality. Although the WNF method visually achieves image quality comparable to the proposed approach with minimal perceptual differences in the RFI-suppressed images, the quantitative analysis of these RFI-suppressed images reveals a distinct performance disparity.

As evidenced by the RMSE metrics in Table IV, the proposed method demonstrates superior fidelity to the reference image. Specifically, both WNF and GSVT-LRSD methods exhibit higher RMSE values than the proposed method, demonstrating that the proposed framework effectively improves interference suppression performance by achieving precise PRFI separation while maintaining the integrity of useful signals. Furthermore, as interference intensity increases, the suppression efficacy of the FNF, RPCA, WNF, and GSVT-LRSD methods deteriorates, evidenced by their increasing RMSE values. In contrast, PCA demonstrates improved suppression capability under stronger interference.

B. Experiment Results of Measured SAR Data

It is well known that Sentinel-1 employs the terrain observation with progressive scans SAR (TOPSAR) mode for data acquisition [48]. This advanced imaging technique, a variant of ScanSAR, operates by cyclically steering the antenna beam across multiple adjacent subswaths to acquire data in bursts, thereby achieving wide-area coverage while maintaining high azimuth resolution [48]. In the measured data experiment, real level-1 SLC images from the Sentinel-1 SAR satellite in IW observation mode are utilized to demonstrate the effectiveness and superiority of the proposed method for RFI suppression. All SLC images are acquired from representative scenarios distributed globally, encompassing various types of land cover to ensure a comprehensive evaluation. In order to quantitatively evaluate PRFI suppression performance in measured data experiments, we utilize the AG metric and the ENL metric to assess the quality of the RFI-suppressed SLC image.

The AG metric quantifies the rate of intensity variation across fine-scale image details [49], representing the spatial rate of density changes in multiple dimensions. This metric serves as an effective indicator of localized contrast and edge sharpness, thereby characterizing the relative clarity of the

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TABLE IV
RMSE RESULTS OF SIMULATED RFI SUPPRESSION EXPERIMENTS

ISR \ Methods	FNF	PCA	RPCA	WNF	GSVT-LRSD	**Proposed**
0dB	0.1064	0.2418	0.1206	0.1579	0.1113	**0.1033**
10dB	0.1119	0.2186	0.2967	0.1637	0.1338	**0.0814**
20dB	0.3351	0.1676	0.9686	0.1751	0.3220	**0.1542**

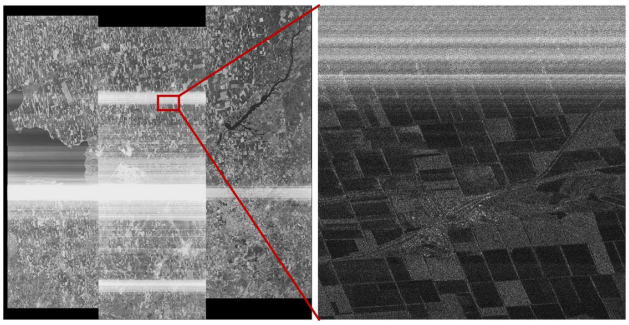


Fig. 6. SAR product image and Scene A.

image. The mathematical formulation of the AG metric is expressed as follows:

$$\mathrm{AG} = \sum_{i=2}^{N_r}\sum_{j=2}^{N_a} \frac{\sqrt{\left(\mathbf{I}_{i,j} - \mathbf{I}_{i,j-1}\right)^2 + \left(\mathbf{I}_{i-1,j} - \mathbf{I}_{i,j}\right)^2}}{\sqrt{2}(N_r - 1)(N_a - 1)} \tag{30}$$

where $\mathbf{I}_{i,j}$ represents the amplitude of the corresponding pixel of the image. Higher values of the AG metric correspond to enhanced edge sharpness and richer structural details, signifying more effective PRFI suppression. Conversely, lower AG values indicate reduced image clarity, reflecting incomplete PRFI suppression.

The ENL serves as a quantitative metric for assessing the homogeneity of uniform regions in SAR imagery [50], characterizing the intensity of multiplicative noise present in the image. As one of the key indicators of SAR image quality, a lower ENL value indicates less multiplicative noise contamination and consequently higher image quality. ENL is mathematically defined as the squared ratio of the mean to the standard deviation of the amplitude of the SAR SLC image, expressed as

$$\mathrm{ENL} = \left(\frac{\mu_{\mathbf{I}}}{\sigma_{\mathbf{I}}}\right)^2 \tag{31}$$

where $\mu_{\mathbf{I}}$ and $\sigma_{\mathbf{I}}$ represent the mean and standard deviation of the amplitude of the SAR SLC image, respectively. The mean value provides a measure of the overall brightness level across the SAR image, while the standard deviation quantifies the degree of intensity variation, serving as an indicator of spatial heterogeneity.

1) Scene A: The data of Scene A is acquired by Sentinel-1 near Rostov in Russia, as illustrated in Fig. 6. The SLC product image reveals severe PRFI contamination, with interference artifacts obscuring extensive areas of the scene. This degree of degradation has rendered the SAR observation of Rostov city effectively inoperative, compromising both interpretability and utility of the acquired data. Due to the extensive data volume of the original SLC image, we selected a representative subset (Scene A) for qualitative and quantitative analysis of the PRFI suppression performance of the proposed algorithm. Spectral analysis of the data reveals that interference in Scene A exhibits comb-spectrum interference characteristics, which is composed of multiple superimposed narrowband PRFI components. This interference structure aligns precisely with the operational assumptions of the proposed suppression methodology, making it particularly suitable for effective mitigation using our approach.

After the PRFI suppression experiments of Scene A, the suppression results of FNF, PCA, RPCA, GSVT-LRSD, and the proposed method are depicted in Fig. 7(b)–(f), respectively. It is evident that all methods demonstrate measurable effectiveness in mitigating the PRFI contamination present in Scene A, albeit with varying degrees of performance. The FNF method effectively removes PRFI artifacts from the SLC image, although a visual observation of Fig. 7(b) reveals a noticeable amplitude reduction in some strong point targets compared with the original SLC image, indicating a potential loss of useful signals. Analysis of Region 1 in Fig. 7(c) and (e) reveals that the PCA and GSVT-LRSD methods achieve partial interference suppression, manifested as interference artifacts remaining to varying degrees. In addition, the observations of Region 2 indicate relatively minor signal loss for these three methods. In contrast, the RPCA approach demonstrates superior interference cancellation capability, as evidenced in Fig. 7(d), but at the cost of significant signal loss and overall reduction in amplitude of the SLC image. As shown in Fig. 7(f), the suppression result of the proposed method reveals substantial improvements in both interference suppression completeness and signal preservation, making the ground targets under the PRFI artifacts clearly visible in Region 1. Moreover, the proposed method exhibits robust performance in both heavily contaminated (Region 1) and minimally affected (Region 2) areas, demonstrating superior capability in achieving precise PRFI separation while preserving useful signals compared with the aforementioned approaches.

To quantitatively assess the PRFI suppression performance of different methods in Scene A, the AG and ENL metrics are used as evaluation metrics of the RFI-suppression images. As evidenced by the quantitative results in Table V, the proposed method achieves better image quality after PRFI suppression compared with methods such as the FNF, RPCA,

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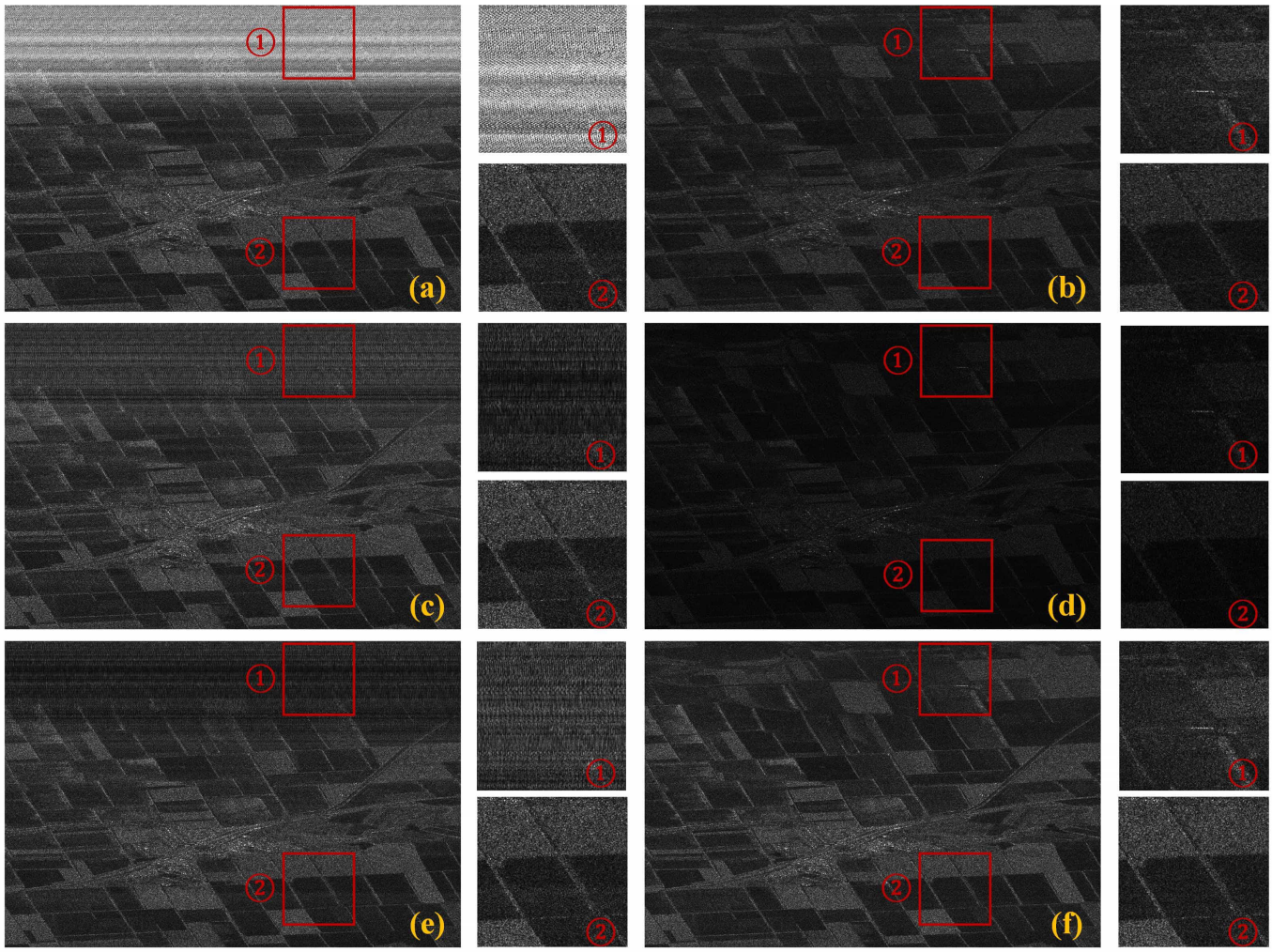


Fig. 7. Experimental results for real SAR data. (a) RFI-affected SAR image of Scene A. Suppression outcome of (b) FNF, (c) PCA, (d) RPCA, (e) GSVT-LRSD, and (f) proposed method.

TABLE V
EVALUATION OF SCENE A AFTER PRFI SUPPRESSION

Metrics \ Methods	FNF	PCA	RPCA	G.L.[1]	W.F.L.[2]
AG	8.21	**11.41**	4.48	10.94	11.07
ENL	2.57	2.34	2.25	2.40	**2.24**

[1] Here G.L. stands for GSVT-LRSD method.
[2] Here W.F.L. stands for the proposed WNF-FGSVT-LRSD method.

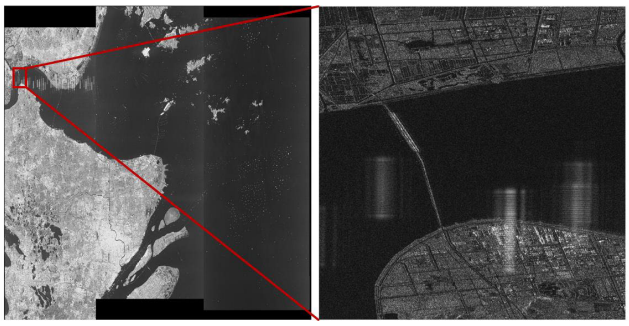


Fig. 8. SAR product image and Scene B.

and GSVT-LRSD, exhibiting higher AG and lower ENL values than these methods. This indicates that the image after interference suppression has clear textures and less multiplicative noise. However, the PCA method yields a slightly higher AG than the proposed approach. This result may be attributed to incomplete PRFI suppression, as is evident in Fig. 7(c), which inflates local gradient measurements and consequently distorts the AG metric.

2) Scene B: The data of Scene B is acquired by Sentinel-1 near Shanghai, China, as illustrated in Fig. 8. As observed, the scene displays relatively low-intensity interference that affects only a limited number of ground targets, presenting a challenging test case for suppression algorithms. Traditional notch filtering approaches face a fundamental tradeoff: overly conservative thresholding selection preserves interference artifacts, while aggressive thresholding selection causes significant signal loss. In contrast, the proposed joint framework simultaneously addresses both intensity characteristics and low-rank sparse properties of narrowband PRFI, enabling superior performance in interference mitigation. Analysis of Scene B reveals narrowband PRFI contamination, indicating its suitability for the proposed method. The comparative results

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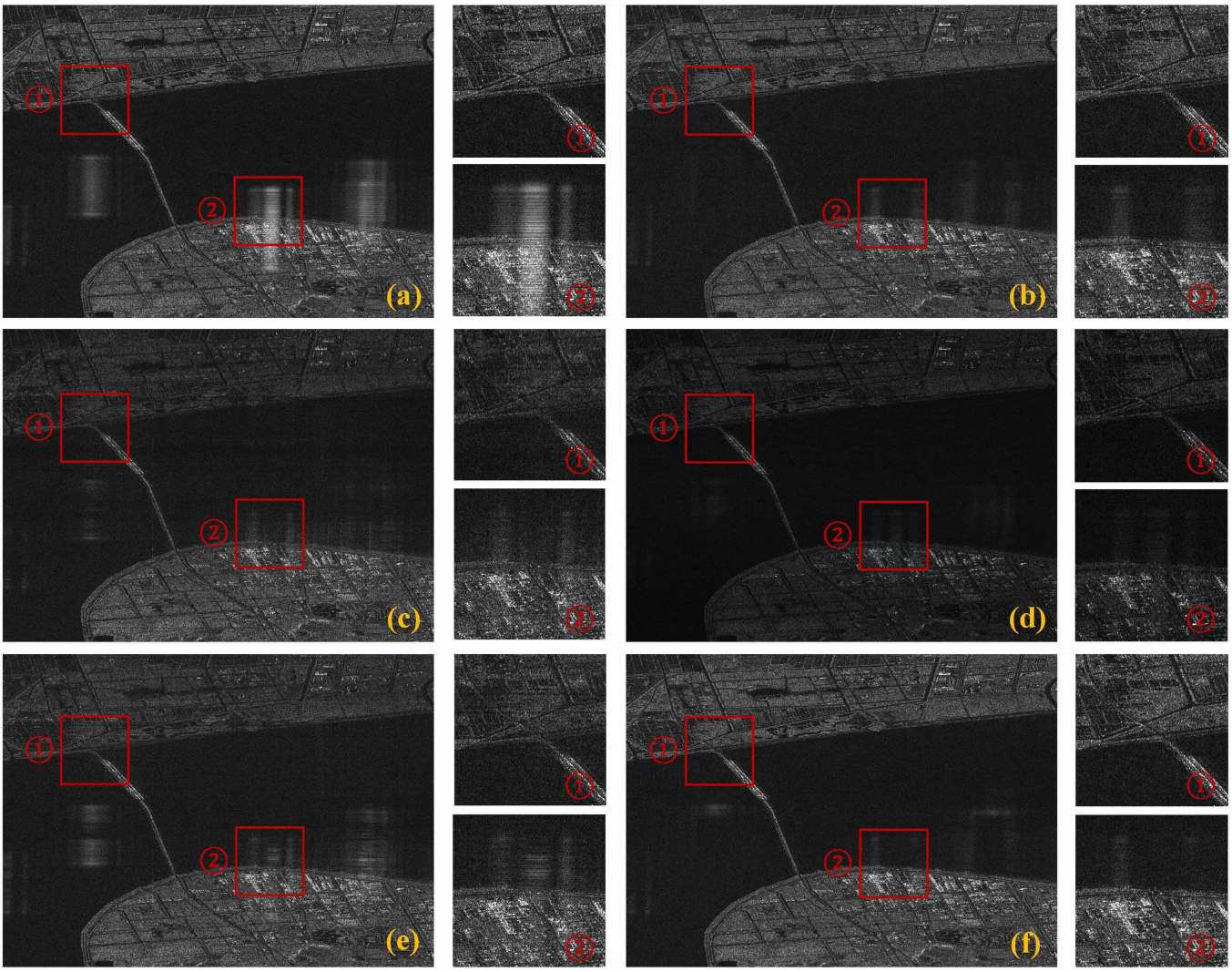


Fig. 9. Experimental results for real SAR data. (a) RFI-affected SAR image of Scene B. Suppression outcome of (b) FNF, (c) PCA, (d) RPCA, (e) GSVT-LRSD, and (f) proposed method.

of various suppression techniques applied to Scene B are presented in Fig. 9.

As depicted in Fig. 9, comparative results of various interference suppression methods are applied to Scene B. Analysis of Region 2 in Fig. 9(c) and (e) reveals limited suppression effectiveness for both PCA and GSVT-LRSD approaches. Furthermore, examination of Region 1 indicates attenuation of strong point targets' scattering intensity, suggesting nonnegligible signal loss. Similar to the experiment results of Scene A, the RPCA method demonstrates effective interference cancellation as shown in Fig. 9(d), but at the cost of amplitude reduction across the whole image. The FNF method shows improved performance, achieving substantial interference mitigation while maintaining signal integrity. However, the proposed method outperforms FNF, as evidenced by significantly reduced residual artifacts in Region 2 of Fig. 9(f), demonstrating superior performance in both interference suppression and signal preservation.

The quantitative results of the evaluation of Scene B after PRFI suppression are presented in Table VI, revealing that the proposed method achieves the highest AG and lowest ENL values among all the evaluated methods. This improvement indicates a significant advancement in RFI-suppressed image quality.

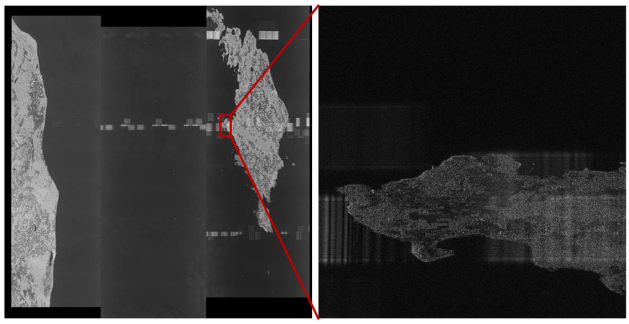


Fig. 10. SAR product image and Scene C.

3) Scene C: The data of Scene C is acquired by Sentinel-1 near Ostergarn island in Denmark, as illustrated in Fig. 10. Similarly, Scene C is contaminated by narrowband PRFI but with a lower energy level than in Scene B. These low-intensity

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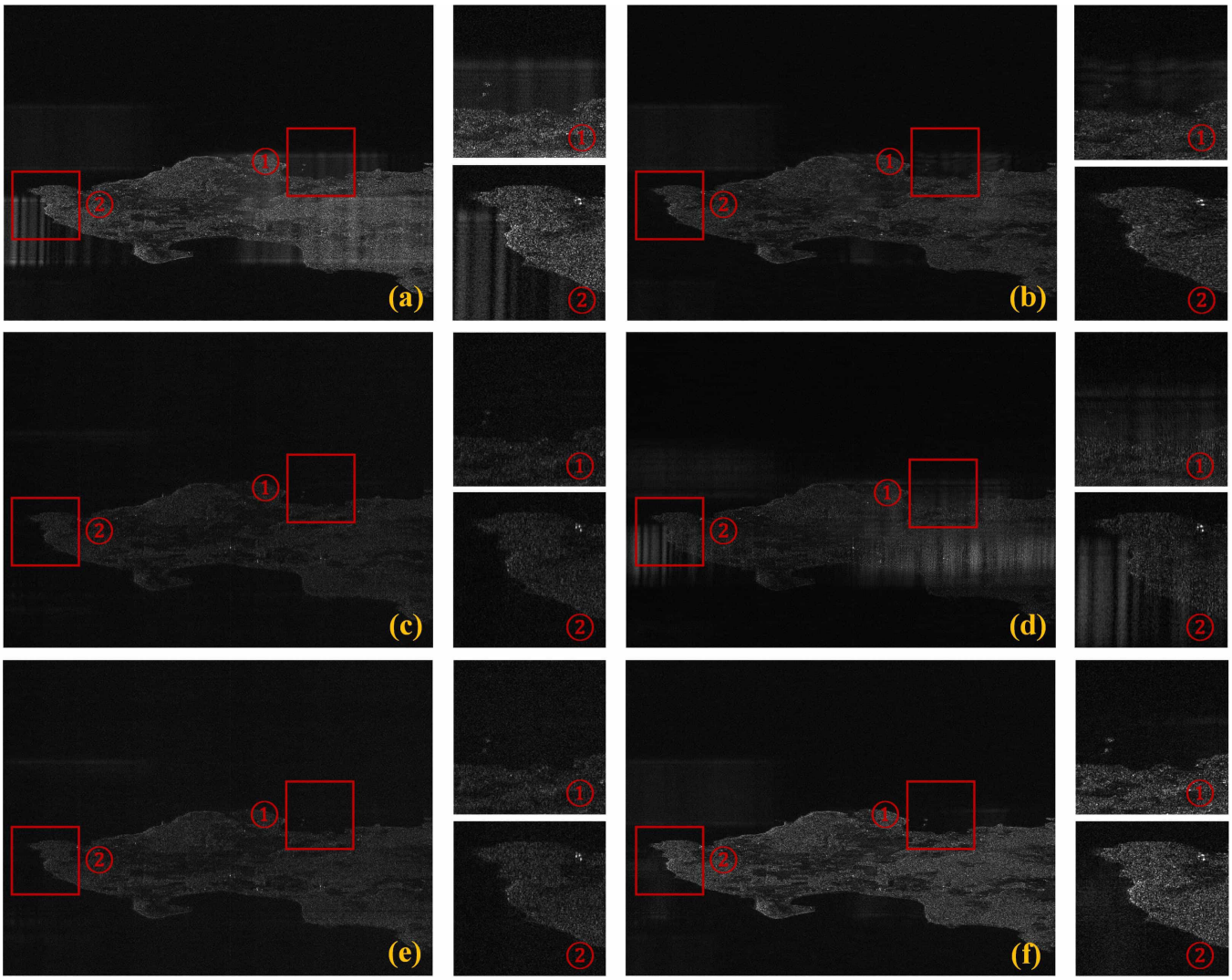


Fig. 11. Experimental results for real SAR data. (a) RFI-affected SAR image of Scene C. Suppression outcome of (b) FNF, (c) PCA, (d) RPCA, (e) GSVT-LRSD, and (f) proposed method.

TABLE VI
EVALUATION OF SCENE B AFTER PRFI SUPPRESSION

Metrics \ Methods	FNF	PCA	RPCA	G.L.[1]	W.F.L.[2]
AG	12.12	10.51	6.21	11.74	**12.57**
ENL	1.39	1.52	1.57	1.57	**1.30**

[1] Here G.L. stands for GSVT-LRSD method.
[2] Here W.F.L. stands for the proposed WNF-FGSVT-LRSD method.

interferences manifest as block-like artifacts that partially obscure targets of land and sea, rendering them barely visible. As observed in Scene B, suppressing weak PRFI presents a challenging task for current methods, mainly due to the inherent tradeoff between interference suppression efficacy and SAR signal preservation. Even with manual fine-tuning of thresholds or hyperparameters, achieving optimal results remains difficult. In contrast, the proposed method effectively balances these competing objectives. Compared with the current methods, the proposed method exhibits relatively outstanding interference suppression performance in Scene C and can achieve the best SAR image quality simultaneously.

The results of the comparative experiment of different RFI suppression methods applied to the SLC data of Scene C are depicted in Fig. 11. It is evident that all methods demonstrate measurable PRFI suppression effectiveness in this scenario. However, as shown in Fig. 11(d), the RPCA approach leaves the most residual interference artifacts. As depicted in Fig. 11(b), (c), and (e), comparative analysis of FNF, PCA, and GSVT-LRSD methods reveals that FNF exhibits relatively higher residual interference, and PCA suffers from severe signal loss, particularly evident in the magnitude of ground targets in Region 1, where ground scenes appear blurred. Among these three methods, GSVT-LRSD achieves the best image quality, although a reduction in the amplitude of ground targets persists compared with the original scene. Finally, comparing the proposed method with GSVT-LRSD, as shown in Fig. 11(e) and (f), our approach demonstrates superior performance. The images of Region 1 and Region 2 show effective interference suppression while maintaining clear land–sea boundaries and preserving the energy of

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TABLE VII
EVALUATION OF SCENE C AFTER PRFI SUPPRESSION

Metrics \ Methods	FNF	PCA	RPCA	G.L.[1]	W.F.L.[2]
AG	4.31	4.03	1.79	4.96	**5.94**
ENL	1.24	2.13	1.10	2.07	**1.02**

[1] Here G.L. stands for GSVT-LRSD method.
[2] Here W.F.L. stands for the proposed WNF-FGSVT-LRSD method.

ground targets, which results in the highest overall image quality among the RFI-suppressed images of the five evaluated methods.

By conducting a quantitative evaluation of the image quality of the processed results in Fig. 11, according to Table VII, it is demonstrated that the proposed method achieves better performance in both metrics compared with the four existing approaches. This indicates a robust PRFI suppression capability of the proposed method in Scene C, consistent with its performance in other scenarios.

In summary, the proposed method demonstrates superior results over current techniques when applied to diverse real-world SAR scenes contaminated by narrowband PRFI of varying intensities and characteristics. Its consistent performance across different conditions highlights both robustness and adaptability of the proposed method.

C. Analysis of the Association Between the Two Steps of the Joint Framework

The experimental results demonstrate that the proposed joint framework achieves superior interference suppression performance compared with the WNF method and FGSVT-LRSD method alone, due to magical chemical reactions between them. First, the WNF method leverages interference intensity characteristics in the wavelet domain, while FGSVT-LRSD exploits low-rank properties in the range–frequency domain. This provides complementary discrimination capabilities between interference and SAR signals. Second, this joint framework enhances robustness to threshold selection, as FGSVT-LRSD can recover residual SAR signal components from the signals extracted by WNF, thereby reducing the sensitivity to threshold selection. Third, WNF preprocessing facilitates the low-rank assumption required by FGSVT-LRSD, creating favorable conditions for subsequent LRSD. The following section provides a detailed analysis of these mechanisms.

The WNF method, as a notch filtering approach, determines its threshold based on the statistical distribution of data values of the range spectrum, fundamentally relying on intensity characteristics for interference separation. In contrast, the LRSD method operates under distinct assumptions of low-rank interference and sparse echoes, as formalized in the LRSD model of (4). By integrating these approaches, the proposed framework simultaneously exploits both high-intensity and low-rank properties of interference, enabling more precise separation. Moreover, the integration of the LRSD-based

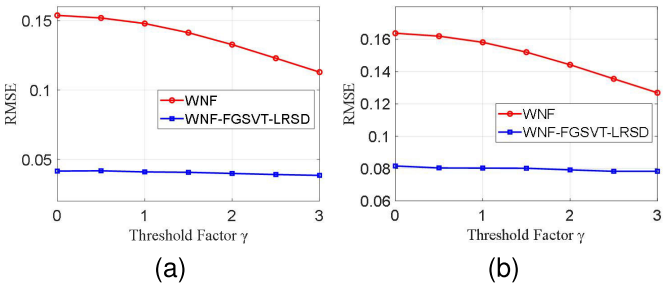


Fig. 12. RMSE variation curve with respect to threshold factor γ. (a) ISR = 0 dB. (b) ISR = 10 dB.

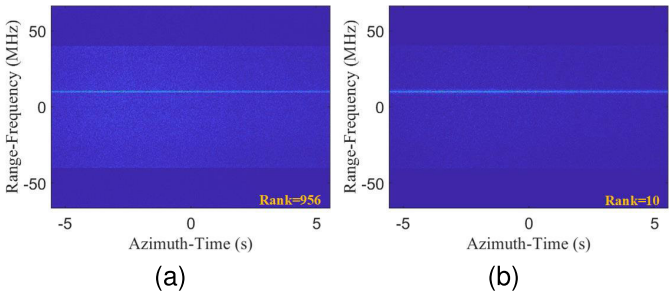


Fig. 13. Range spectrum and its matrix rank (a) before and (b) after WNF.

method within the proposed framework significantly reduces the sensitivity to threshold selection in the notch filtering part of the proposed method. As demonstrated in Fig. 12 using simulated data, the RMSE between WNF-processed and reference images varies substantially with threshold factor γ, indicating a strong performance dependence on precise threshold tuning. On the contrary, the joint framework exhibits minimal RMSE variation across the same range of γ, demonstrating robust performance despite changes in threshold. This threshold insensitivity effectively mitigates signal loss caused by suboptimal threshold selection, which is a critical limitation of notch filtering approaches.

Beyond the aforementioned synergies, WNF processing in the joint framework facilitates satisfaction of the low-rank and sparsity conditions required by LRSD-based methods. By using the same SAR and RFI parameters as in the simulation experiment with an ISR of -10 dB, we generated range spectrum images before and after WNF processing, as illustrated in Fig. 13(a) and (b). The SVT operation is applied to calculate the matrix ranks of the range spectrum, with a threshold of $\kappa = \max(\mathbf{U}_s)/5$, where $\mathbf{U}_s$ is the singular value matrix of $\mathbf{S}$. The results demonstrate that WNF processing reduces the rank of matrix $\mathbf{S}$, satisfying the low-rank and sparsity assumptions better. This rank reduction also accelerates the rSVD computations in the FGSVT-LRSD algorithm. As shown in Table VIII, under identical matrix dimensions and hardware conditions, the computation time of FGSVT-LRSD decreases significantly after WNF processing.

These experimental results demonstrate the mutually reinforcing relationship between WNF processing and FGSVT-LRSD algorithms, providing statistical evidence for the advantages of the proposed method over the compared methods.

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TABLE VIII
TIME CONSUMPTION OF FGSVT-LRSD ALGORITHM

	Before WNF	After WNF
Running Time (s)	584.3	39.7

D. Computation Complexity Analysis

The main computational load of the proposed method originates from the iterative solution of (15), where SVD operations dominate the computational cost. Each SVD computation requires $\mathcal{O}(mn \cdot \min(m,n))$ floating-point operations [51], making it the most resource-intensive component of the optimization process. Based on Table I, we can avoid the expensive computation by calculating SVD on a smaller matrix. As mentioned in Section III, the RFI-affected range spectrum $\mathbf{S}_a$ is of size N_r along the range–frequency axis and N_a along the azimuth–time axis, and its truncate spectrum is $\mathbf{S}_T \in \mathbb{C}^{M\times N_a}$ where $M \geq N_a$. Since the WNF processing preserves matrix dimensions, the input matrix of the FGSVT-LRSD algorithm maintains the same size as the truncated spectrum matrix $\mathbf{S}_T$, denoted as $\mathbf{X} \in \mathbb{C}^{M\times N_a}$. The SVD of matrix $\mathbf{X}$ has a computational complexity of $\mathcal{O}(MN_a^2)$. However, the proposed fast GSVT operator iterates the calculation of rSVD of matrix $\mathbf{X}$, finding a rank-K approximation ($K < N_a$) of $\mathbf{X}$ as $\hat{\mathbf{X}} \in \mathbb{C}^{M\times K}$. Then, the complexity of SVD reduces to $\mathcal{O}(MK^2)$. Therefore, when $K \ll N_a$, by avoiding full-rank SVD computation, the computation time of SVD can be significantly reduced. In fact, the narrowband PRFI suppression task address in the aforementioned section satisfies the condition $K \ll N_a$. Thus, when calculating GSVT, using random SVD instead of full-rank SVD can greatly reduce the calculation time of the algorithm.

V. CONCLUSION

Most of the current RFI suppression methods emphasize on suppressing RFI in the level-0 echo domain, while few of them conduct RFI suppression in the level-1 SLC image domain. However, the level-0 raw data are difficult to access and process for most end-users. In this article, a joint framework of the WNF and FGSVT-LRSD method is proposed to suppress narrowband PRFI in SAR level-1 SLC images. The proposed framework takes advantage of both nonparametric and semiparametric methods, which leverage intensity and low-rank characteristics of narrowband PRFI at the same time. The synergistic integration of WNF processing and the FGSVT-LRSD algorithm within the proposed joint framework reduces the sensitivity to threshold selection in the WNF process while simultaneously ensuring the fulfillment of low-rank and sparsity assumptions required for FGSVT-LRSD. In addition, the computational complexity of GSVT-LRSD decreases greatly by using rSVD, which mitigates the calculation burden of LRSD-based methods. Finally, the experimental results using GF-3 simulated SAR data and Sentinel-1 measured SAR data demonstrate that the proposed method is capable of suppressing narrowband PRFI successfully without introducing large distortion to SLC images. Notably, the proposed method is based on a low-rank assumption for narrowband RFI in the range–frequency domain. Consequently, the effectiveness of the proposed method may degrade when dealing with wideband interference or in more complex electromagnetic scenarios. Addressing this limitation will be the focus of our future research efforts.

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